

UE25CS242A: Operating Systems (4-0-0-4-4)

This course focuses on fundamental operating systems concepts including various algorithms and tradeoffs for efficient management of resources such as CPU, Memory, Storage and I/ O. This course requires the student to have a desirable knowledge of Data Structures and its Applications.

Course Objectives:

- Focus on fundamental Operating System concepts.
- Provide an understanding of various components of the Operating System (OS).
- Delve deeper into various algorithms and associated trade-offs for efficient resource management such as process, disk, and memory management.
- Introduce design principles and trade-offs in the design of Operating Systems.

Course Outcomes:

At the end of the course, the student will be able to:

- Understand the principles and modules of Operating Systems.
- Understand the design of various algorithms for scheduling and their relative performance.
- Understand the concept of Deadlocks that typically occur in OS. Deadlocks - Avoidance and Detection.
- Implement Operating Systems Concepts related to process management, Concurrent processes, Threads and Memory Management.

Course Content:

Unit 1 : Introduction and Process Management – 14 hours

What Operating Systems Do, Computer-System Organization, Computer-System Architecture, OperatingSystem Structure & Operations, Operating-System Services; **Process Concept:** Process in Memory, Process State, Process Control Block (PCB), Process Creation and Termination, System Calls for Process Management, Life Cycle of I/O Requests. **CPU Scheduling:** Basic Concepts, Scheduling Criteria, Scheduling Algorithms: First-Come First-Served (FCFS), Shortest-Job-First (SJF), Priority Scheduling, Round-Robin Scheduling, Multilevel Queue Scheduling, Multilevel Feedback Queue Scheduling; **Introduction to Shell Programming:** Variables, Control Flow, Shell Scripts, Scheduler – cron.

Unit 2 : IPC, Threads and Concurrency – 14 hours

IPC: Shared Memory, Message Passing, Pipes – Named and Ordinary, System Calls for Shared Memory, Pipes and FIFOs, Linux Domain Sockets; **Introduction to Threads:** Types of Threads, Multicore Programming, Multithreading Models, Thread Control Block (TCB), Thread Creation, Thread Scheduling, Thread Libraries, POSIX Threads (Pthreads); **Multicore Kernel Concepts:** Symmetric Multiprocessing (SMP), Kernel Parallelism, CPU Affinity, Load Balancing, Multicore Scheduling; **Mutual Exclusion and Synchronization:** Software Approaches, Principles of Concurrency, Hardware Support, Mutex Locks, Semaphores; Scalable Synchronization: Spinlocks, Reader-Writer Locks, Lock Contention; Classic Synchronization Problems; **Deadlocks:** Principles of Deadlock, Deadlock Characterization, Deadlock Avoidance; Signals and Signal Handling.

Unit 3: Memory Management – 14 hours

Main Memory: Hardware and Control Structures, OS Support, Address Translation, Dynamic Loading, Dynamic Linking and Shared Libraries, Memory-Mapped Files and mmap(), Swapping, Fragmentation, Segmentation, Paging, Structure of Page Tables, Translation Lookaside Buffer (TLB), TLB Hits and Misses, Effective Access Time (EAT); **Virtual Memory:** Demand Paging, Page-Fault Handling, Copy-on-Write, Page Replacement,

Allocation of Frames, Thrashing, NUMA (Non-Uniform Memory Access) Basics and Memory Locality, Clock Page Replacement.

Unit 4 : File Management, Storage Management and Security – 14 hours

File Concept: Access Methods, Directory, Driver Development, Flash Drives, File-System sharing, File system protection, System calls to retrieve file attributes, file types, file operations, reading directories, File system implementation, Directory implementation, allocation methods, Free-Space Management, Efficiency and Performance, **Mass storage structure**, Disk Structure, Disk scheduling, Disk Management, RAID Structure, OS Security, Access Matrix, Implementation of the Access Matrix, Access Control, Revocation of Access Rights; Shell programming – awk, sed.

Tools/Languages/OS : C, Linux/Unix OS for system call implementation.

Text Book(s):

1. “Operating System Concepts”, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne 10th Edition, John Wiley & Sons, Global Edition, 2023.
2. “Advanced Programming in the Unix Environment”, Richard Stevens and Stephen A Rago, Pearson, 3rd edition, 2017.

Reference Book(s):

1. “Operating Systems, Internals and Design Principles”, William Stallings, 9th Edition, Pearson, 2018.
2. “Modern Operating Systems”, Andrew S Tanenbaum, 3rd edition, Pearson, 2007.
3. “Learning the bash shell”, Cameron Newham, 3rd edition, O’Reilly, 2005